THE AFFINE STRATIFICATION NUMBER UNDER A POSITIVE WEIGHTING OF A SURJECTIVE QUASI-FINITE MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 23089-23118, immediately following the statement at 23079-23087.

Editorial note. The footnote on 'separated' (source line 23082: it suffices that the diagonal of Y is affine) is retained verbatim inside the counted statement span.

Editorial note. The same-chapter pointer family (the max-int lemmas and the weighting base-change lemma) consists of line-indexed stubs into the pinned source.

Allowed standard prerequisites (editorial).

- Separated target makes quasi-finite morphisms affine. Morphisms, Lemma morphisms-lemma-affine-permanence, cited at proof lines 23092-23093. Gives that X -> Y is affine once Y is separated, the standing input for every 'affine over X is affine' step.
- Maximum of the weighted integral and finiteness over its stratum. Lemmas lemma-max-int-w (line 23095) and lemma-max-int-finite (lines 23098-23099) of this chapter. Supplies the induction parameter d and that f^{-1}(Y_d) -> Y_d is finite.
- Affineness of the maximum stratum. Lemma lemma-affineness-of-large-open of this chapter, cited at proof lines 23099-23102. Y_d meets every affine open W of Y affinely, hence Y_d -> Y is an affine morphism of schemes.
- Affineness from an affine surjective cover. Limits, Lemma limits-lemma-affine, cited at proof lines 23108-23109. Y_d is affine because it is surjected upon by the affine scheme f^{-1}(Y_d).
- Weightings are stable under base change. Lemma lemma-weighting-base-change of this chapter, cited at proof lines 23119-23121. Lets the induction start again on the reduced closed part Y' with the restricted weighting w'.

Source reference. Original lemma 0F3R: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 22966–22966. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0F3L: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 22749–22749. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0F3K: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 22731–22731. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted

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proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0F3B: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 22173–22173. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01ZT: focused-seeds/geometry/S9-morph-asn-weighting/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01SG: focused-seeds/geometry/S9-morph-asn-weighting/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Proposition 0.1. *Let $f : X \to Y$ be a surjective quasi-finite morphism of schemes. Let $w : X \to \mathbf{Z}_{>0}$ be a positive weighting of f. Assume X affine and Y separated*[1] *and nonempty. Then the affine stratification number of Y is at most the number of distinct values of $\int_f w$ minus 1.*

Proof. Note that since Y is separated, the morphism $X \to Y$ is affine (Morphisms, Lemma 01SG). The function $\int_f w$ attains its maximum d by Lemma 0F3K. We will use induction on d. Consider the open subscheme $Y_d = \{y \in Y \mid (\int_f w)(y) = d\}$ of Y and recall that $f^{-1}(Y_d) \to Y_d$ is finite, see Lemma 0F3L. By Lemma 0F3R for every affine open $W \subset Y$ we have that $Y_d \cap W$ is affine (this uses that $W \times_Y X$ is affine, being affine over X). Hence $Y_d \to Y$ is an affine morphism of schemes. We conclude that $f^{-1}(Y_d) = Y_d \times_Y X$ is an affine scheme being affine over X. Then $f^{-1}(Y_d) \to Y_d$ is surjective and hence Y_d is affine by Limits, Lemma 01ZT. If $Y_d = Y$, then the affine stratification number of Y is 0 and we conclude the lemma is true. If not, then let $Y' = Y \setminus Y_d$ viewed as a reduced closed subscheme of Y. Set $X' = Y' \times_Y X$. Since X' is closed in X it is affine. Since Y' is closed in Y it is separated. The morphism $f' : X' \to Y'$ is surjective and w induces a weighting w' of f' (see Lemma 0F3B) and $\int_{f'} w'$ is the restriction of $\int_f w$ to Y'. By induction Y' has an affine stratification of length $\leq$ the number of distinct values of $\int_{f'} w'$ minus 1 and the proof is complete. □

[1] It suffices if the diagonal of Y is affine.